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Full-face treatment with onabotulinumtoxinA: Results from a single-center study

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Abstract

Background: Botulinum toxin type A injection is a common aesthetic treatment that can be used alone or in combination to rejuvenate the entire face and neck.

Objectives: To assess the safety and efficacy of a full-face approach with botulinum toxin type A.

Methods: This was a single-center, retrospective review of data from 189 patients receiving full-face and neck treatment with onabotulinumtoxinA in the following areas: glabellar lines, crow's feet lines, forehead lines, bunny lines, sagging nasal tip, gummy smile, marionette lines, mentalis, masseter, and lateral/anterior platysma.

Results: The mean age was 56.6 ± 9.2 years, and 94.7% were women (n = 179). Patients received a total of 394 full-face treatments (mean: 2.1 per person; range 1-7). The mean quantity of onabotulinumtoxinA injected was 113.8 ± 17.6 units per treatment (range: 50-180 units). The most frequently injected areas were the glabella/crow's feet/forehead (n = 383; 97.2%), platysma (n = 378; 95.9%), bunny lines (n = 283; 71.8%), and chin (n = 120; 30.5%). Six patients (3.2%) experienced complications: five cases of unwanted diffusion from the mentalis into the depressor labii inferioris and one case of right eyelid ptosis after injection in the upper third. All were minor and none appeared to be related to overall dose.

Conclusions: Full-face and neck treatment with onabotulinumtoxinA can be performed without substantial safety concerns, often leading to excellent results. This approach may become increasingly prevalent in the coming years.

KEYWORDS

botulinum toxin type A, full-face treatment, onabotulinumtoxinA

1 | INTRODUCTION

The face and neck form a morphofunctional unit in which the mimic muscles act to modify its shape. These muscles attach partly to the skin, and hence when they contract, they cause the skin to move. As people age, signs of the action of mimic muscles become increasingly evident, particularly in the form of wrinkles and furrows. These often remain visible even in the absence of active movement when the face is static. In addition, the elasticity of the skin declines with age, which may make underlying muscle action more visible.

Botulinum toxin type A (BoNTA) blocks cholinergic synapse transmission, thus reducing muscular activity.[1] It has long been known that BoNTA can be used to improve the aesthetic appearance of facial lines,[2] with clinical effects becoming evident within a few days and reaching maximum effect within 2 weeks.[1] Results typically last for ≥4 months.[3,4]

Figures from the American Society of Plastic Surgeons suggest that BoNTA injection was by far the most common aesthetic treatment in the US in 2018, with more than 7 million procedures performed.[5] To put that into context, the second most common

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aesthetic treatment was soft-tissue filler injection, which accounted for fewer than 3 million procedures. Furthermore, the total number of nonsurgical treatments like BoNTA and fillers dwarfed surgical aesthetic procedures by a factor of almost ten.[5] This may reflect a preference among patients for the minimal invasiveness and shortened recovery time of nonsurgical treatments.

OnabotulinumtoxinA is an approved BoNTA formulation with proven long-term safety, efficacy and patient satisfaction in the treatment of lines across different areas of the face.[6-11] Although the currently approved indications in the European Union relate to moderate-to-severe glabellar, crow's feet, and forehead lines,[12] off-label use in clinical practice often extends well beyond these areas. For example, a recent Italian consensus provided recommendations on the use of onabotulinumtoxinA for glabellar lines, crow's feet lines, forehead lines, brow lifting/shaping, hypertrophy of the lower eyelid orbicularis oculi, bunny lines, sagging nasal tip, gummy smile, hypertrophy of the masseter muscle, perioral lines, marionette lines, mentalis muscle hypertonia, and hypertonic platysma bands.[13]

This wide range of viable treatment areas creates the possibility of "full-face" treatment with onabotulinumtoxinA. To obtain an anti-aging effect on the face and neck using BoNTA, based on a full-face approach, it is essential to evaluate the direct and indirect effects of all the muscles involved in creating the visible effects at surface level. Practitioners must act in a targeted way to rebalance the tensile forces and stimulate an uplift effect through modulation of the action of the various muscle groups involved in forming facial expressions. Thus, for example, we can reduce the activity of the platysma, which has a depressive activity through its mandibular insertion. The present analysis provides data from a group of patients treated with onabotulinumtoxinA for aesthetic purposes, based on a full-face approach.

2 | MATERIALS AND METHODS

2.1 | Study design and participants

This was a retrospective review of data from patients receiving full-face treatment with onabotulinumtoxinA (Vistabex®, Allergan) at a single center between April 2016 and March 2019. It was conducted in accordance with the Declaration of Helsinki. All subjects provided written informed consent.

Eligible patients were adult males and females wishing to receive nonsurgical, aesthetic facial treatment using onabotulinumtoxinA. Key exclusion criteria were previous treatment within the past 3 months, allergy to any of the studied products, relevant comorbidities that might interfere with treatment, and pregnancy or breastfeeding.

2.2 | Procedures

All patients were injected with onabotulinumtoxinA for the treatment of one or more of the following: glabellar lines, crow's feet lines, forehead lines, bunny lines, sagging nasal tip, gummy smile, marionette lines, mentalis, masseter, and lateral/anterior platysma. With regard to dilution, 50 unit vials of onabotulinumtoxinA were reconstituted with 1.25 mL of 0.9% saline to treat glabellar lines, crow's feet lines, forehead lines, bunny lines, sagging nasal tip, gummy smile, marionette lines, and mentalis; and with 1.50 mL of 0.9% saline to treat the masseter and lateral/anterior platysma.

Injection points are shown in Figure 1, with facial anatomy painted onto the skin surface. The use of body painting has been proposed as a useful tool in the teaching of BoNTA,[14] although it is not a technique that can be applied in routine practice. Details of the quantity of onabotulinumtoxinA and injection depth in each area are provided in Table 1.

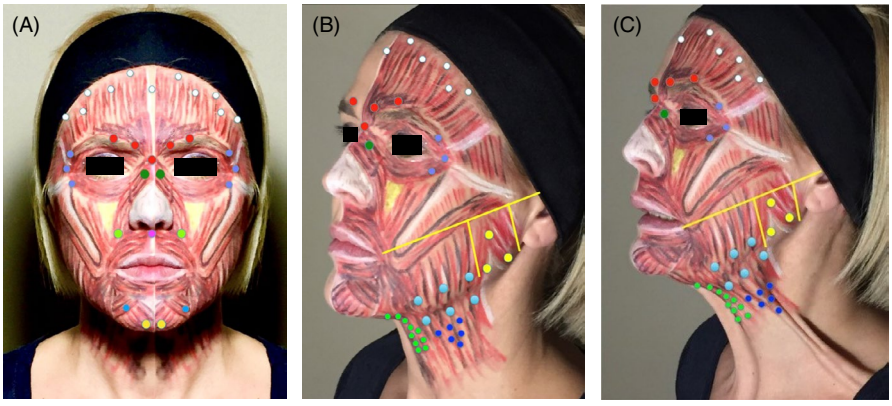


FIGURE 1 OnabotulinumtoxinA injection points. In parts A–C: *white:* 6-20 units in multipoint to modulate frontalis action; *red:* 20-30 units in five points into the procerus and corrugator supercilii; *purple:* 12-17 units in each side into the orbicularis oculi; *dark green:* 4-12 units into the nasalis. In part A only: *pale green:* 2-4 units in one point across the zygomaticus minor, levator labii superioris and levator labii superioris alaeque nasi; *pink:* 2 units into the depressor nasalis; *pale blue:* 2 units into the depressor labii inferioris; *yellow:* 4-6 units into the mentalis. In parts B and C only: *yellow:* 20-35 units in a double plain into the masseter muscle on each side; *pale blue:* 14-16 units in six points 1 cm above and below the mandibular line on each side; *dark blue:* 4-6 units in the lateral platysma band in small intradermal points on each side; *pale green:* 10 units in the submental and anterior area of the neck in intradermal micropoint

TABLE 1 Quantity of onabotulinumtoxinA injected in individual areas of the face and neck

Area treated	Dosages	Preferred injection level
Glabellar lines	20-30 units	Intramuscular
Crow's feet lines	24-34 units	Intracutaneous
Forehead lines	6-20 units	Intramuscular or intracutaneous
Bunny lines	4-12 units	Intramuscular
Sagging nasal tip	2 units	Intramuscular
Gummy smile	4-8 units	Intramuscular
Marionette lines	4 units	Intramuscular
Mentalis muscle	4-6 units	Intramuscular
Masseter muscle	40-70 units	Intramuscular
Lateral and anterior platysma	50-60 units	Intracutaneous

At each treatment session, static and dynamic photographs were taken from various angles prior to treatment with onabotulinumtoxinA, at 2-3 weeks post-treatment (with touch-up allowed when indicated), and again at 4-5 weeks to evaluate the final result. For each patient, a graphic chart was created, in which the sites and dosages injected during the treatment session were recorded, as well as any touch-ups required.

In patients undergoing more than one session of treatment, the time between sessions was typically 4-6 months.

2.3 | Assessments

Baseline characteristics (age and sex) were recorded for all patients. Data collected included the facial areas treated with onabotulinumtoxinA and the number of units used during each treatment session for each patient. Complications were documented from patient reports and through comparison of before-and-after photographs.

2.4 | Statistical analysis

Descriptive statistics are provided throughout, including mean, standard deviation and range for continuous variables, and frequency and percentage for categorical variables.

3 | RESULTS

A total of 189 patients were included in the analysis, of whom 179 (94.7%) were female and 10 (5.3%) were male (Table 2). The mean age was 56.6 ± 9.2 years (range: 34-82 years). Included patients received a total of 394 full-face treatments, equivalent to a mean of 2.1 per person (range 1-7). Ninety-eight patients (51.9%) received one session of treatment, and the remaining 91 (48.1%) received two or more sessions. The mean quantity of onabotulinumtoxinA injected across each of the 394 treatment sessions was 113.8 ± 17.6 units (range: 50-180 units). This includes touch-ups (typically 6-35 units, if required). The overall mean quantity of onabotulinumtoxinA per treatment session was slightly higher in men than women (123.5 ± 9.5 vs 113.1 ± 17.9 units, respectively). As shown in Table 2, the mean amounts used were relatively consistent across each treatment session, with a slight trend toward more units being used in later sessions.

TABLE 2 Patient characteristics and treatments

Characteristic	Patients (N = 189)
Age (y), mean (SD, range)	56.6 (9.2, 34-82)
Sex, n (%)	
Female	179 (94.7)
Male	10 (5.3)
Number of full-face treatment sessions, n (%)	
1	98 (51.9)
2	40 (21.2)
3	19 (10.1)
4	13 (6.9)
5	10 (5.3)
6	6 (3.2)
7	3 (1.6)
OnabotulinumtoxinA injected per treatment (units), mean (SD, range)	
All sessions (N = 394)	110.9 (16.0, 50-160)
Session 1 (N = 189)	113.8 (17.6, 50-180)
Session 2 (N = 91)	113.6 (16.4, 50-160)
Session 3 (N = 51)	118.7 (20.5, 50-180)
Session 4 (N = 32)	117.0 (19.7, 50-170)
Session 5 (N = 19)	117.5 (22.3, 53-175)
Session 6 (N = 9)	123.1 (20.5, 100-170)
Session 7 (N = 3)	124.7 (5.0, 120-130)

Abbreviation: SD, standard deviation.

Among the 394 full-face treatments given, the most commonly injected areas were the glabella/crow's feet/forehead (n = 383; 97.2%), platysma (n = 378; 95.9%), bunny lines (n = 283; 71.8%), and chin (n = 120; 30.5%). Other injected areas are shown in Figure 2.

Sample images showing patients before and after full-face treatment with onabotulinumtoxinA are provided in Figures 3-6

Complications were recorded in 6 patients (3.2%). Of these, 5 patients (2.6%) experienced unwanted onabotulinumtoxinA in one of two depressor labii inferioris (DLI) muscles, most likely due to diffusion from the mentalis; these were treated by touch-up of the

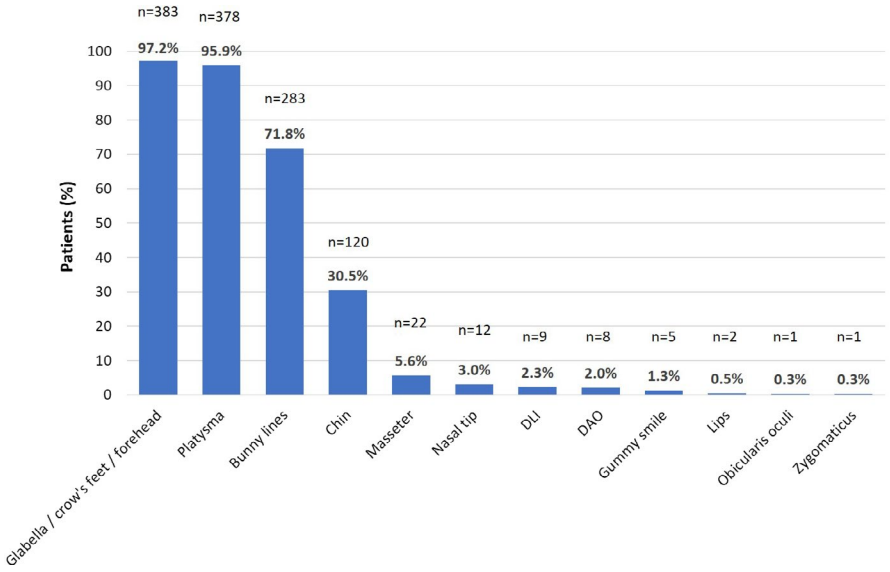


FIGURE 2 Areas of the face and neck injected. Data are percentages of all treatments (N = 394). "Obicularis oculi" indicates treatment of this area without treatment of other key areas of the upper third. DAO, depressor anguli oris; DLI, depressor labii inferioris



FIGURE 3 Full-face treatment with onabotulinumtoxinA. Static and dynamic photographs of a 59-y-old woman before treatment (A–E) and at 4 wk after treatment (F–J) with a total of 120 units of onabotulinumtoxinA in the glabella, crow's feet, forehead, bunny lines, mentalis and platysma. The impact on the platysma (pretreatment, image D; post-treatment, image I) and mentalis (pretreatment, image E; post-treatment, image J) is particularly notable

contralateral DLI with 2 units of onabotulinumtoxinA. One patient (0.5%) experienced right eyelid ptosis after injection in the upper third of the face. To mitigate this, 9 units of onabotulinumtoxinA were injected in the orbicularis oculi to stimulate the tone of the eyelid lift, and apraclonidine eye drops were prescribed for a few days. None of the individuals who experienced complications received overall quantities of onabotulinumtoxinA at the top end of the range, and hence they did not appear to be related to overall dose.

4 | DISCUSSION

This analysis demonstrates that full-face and neck treatment with onabotulinumtoxinA can be performed without substantial safety concerns when correctly administered by a practitioner with sufficient understanding of facial anatomy. Indeed, complications were recorded in only 3.2% of patients; all were most likely related to diffusion (although, in some cases, it is also possible that variations



FIGURE 4 Full-face treatment with onabotulinumtoxinA. Static and dynamic photographs of a 66-y-old woman before treatment (A–G) and at 4 wk after treatment (H–N) with a total of 120 units of onabotulinumtoxinA in the glabella, crow's feet, forehead, bunny lines, mentalis and platysma. The impact on the platysma (pretreatment, images C and G; post-treatment, images J and N) is particularly notable



FIGURE 5 Full-face treatment with onabotulinumtoxinA. Static and dynamic photographs of a 52-y-old woman before treatment (A–D) and at 2 wk after treatment (E–H) with a total of 110 units of onabotulinumtoxinA in the glabella, crow's feet, forehead, bunny lines, and platysma. The impact on the platysma (pretreatment, image C; post-treatment, image G) is particularly notable. The result is a younger looking patient, but with preservation of natural expressiveness

in patient anatomy may have led to direct injection of BoNTA into a different muscle from that intended). Only one patient (0.5%) experienced eyelid ptosis, which compares favorably with rates of around 2% observed in a meta-analysis of onabotulinumtoxinA trials

in the treatment of glabellar lines.[6] The frequency of diffusion-related complications was also favorable compared with overall rates observed across three different BoNTA formulations in a systematic review.[15]

The approach taken in this group of patients reflects the current trend within facial aesthetic medicine, which is moving away from treatment of specific modules, toward a more holistic, full-face approach. A number of other studies have demonstrated the efficacy and safety of this paradigm, based on evaluations of full-face rejuvenation with injectable treatments.[16-24] However, most were based primarily on fillers, and few studies have assessed the use of a full-face approach with BoNTA.

The present data were based on a mean of around 110 units of onabotulinumtoxinA per treatment. Some may consider this to be high. However, to put it into context, a 2013 study of full-face treatment with BoNTA randomized 90 patients into three groups based on the quantities injected: 120-165 units, 166-205 units, or 206-250 units.[19,20] Thus, all three groups received higher quantities of BoNTA than the mean in the current analysis, and yet the complication rate did not increase with increasing overall quantity of BoNTA injected.[19] Similarly, in our analysis, complications did not appear to be related to the total amount of BoNTA injected. Hence, it appears that the use of BoNTA across several areas of the face and neck in the same session is safe, as long as recommended doses are not substantially exceeded within each treatment area. The doses used in specific areas of the face and neck in the present work were largely in line with international recommendations,[25] as well as a recent Italian consensus.[13] The "microbotox" technique[26]—in which multiple microdroplets of diluted onabotulinumtoxinA are injected into the dermis or the interface between the dermis and the superficial muscle layer—can also be applied across the whole face.

For many patients, treatment with BoNTA may be considered as a starting point, to be followed by hyaluronic acid filler or other treatments, as appropriate. For other patients—such as those who do not want a volume change but instead prefer a more "natural" anti-aging treatment—BoNTA may be used alone.

As shown in Figures 3-6, treatment can achieve a new balance to the face with clear anti-aging effects. Many patients reported being highly satisfied with outcomes, although efficacy was not objectively assessed. However, to evaluate the results of treatment, we always take static and dynamic pre- and post-treatment photographs from various angles. This aligns with current Italian and international recommendations, which state that such photographs should always be captured, ideally based on standardized lighting, constant distance between camera and subject, and appropriate patient and camera positioning.[13,27,28]

The most commonly injected areas were the glabella/crow's feet/forehead, platysma, bunny lines, and chin. Using a full-face treatment approach, we were able to achieve a number of benefits, based on personalized treatment plans. These included wrinkle reduction in the upper third, reshaping of the mandibular profile, improvement of labial commissures, amelioration of marionette lines with redefinition of the jawline, and also facial slimming effects (if required). The mimic muscles play a major role in expressing emotions, and hence the use of BoNTA as a neuromodulator across the entire face and neck can create a new and more aesthetically pleasing equilibrium and a new means of emotional communication. Furthermore, modulation of the contractile activity of mimic muscles can have a preventive effect by reducing the future development of wrinkles from prolonged contractions.[29] There may also be improvements in skin quality with long-term BoNTA treatment.[29]

Although the currently approved indications for onabotulinumtoxinA are in the upper face (glabella/crow's feet/forehead), it can also be successfully deployed off-label in the lower face and neck. In the present analysis, almost all patients were treated in the platysma and around a third in the chin. Examples of the success of these treatments are shown in Figures 3-6. Furthermore, local and international consensus recommendations now place increasing focus on these areas.[13,26] A recent analysis of 161 patients treated in the lower face with BoNTA—based on two intramuscular injections into the mentalis and two horizontal lines of superficial injections above and below the mandible—demonstrated improved lower facial contouring, as well as relaxation of horizontal lines below the lateral mandibular border and vertical smile lines lateral to the oral commissures and melomental folds.[30] Although the majority of patients seeking BoNTA treatment are women, many male patients can particularly benefit from BoNTA injection in the lower face. Compared with women, key treatment goals in men are less likely to involve the reduction of lines but may instead relate to reducing an overly masculine appearance to the mandible, improved chin contouring, and reduction of platysmal bands.[31]

Half of the present cohort were treated on two or more occasions (at least 4-6 months apart) and almost 20% had undergone four or more treatment sessions by the time of database lock. There is no reason to believe that safety, efficacy, or patient satisfaction is likely to decline over time. Indeed, in a recent long-term analysis of 194 individuals treated with onabotulinumtoxinA for glabellar lines for a mean of 9.1 years, 92.3% reported being mostly or very satisfied and 89.7% said they looked younger.[9]

We should acknowledge the key limitation of the present work: It was a retrospective analysis of data from a single center. Prospective, multicenter studies of a full-face approach to aesthetic treatment with BoNTA would of course be valuable.

5 | CONCLUSIONS

This analysis showed that minimally invasive, full-face and neck treatment with onabotulinumtoxinA—using mean quantities of more than 100 units per treatment session—can be performed with minimal complications and potentially excellent results. Informal assessments of outcomes based on before-and-after photographs, as well as anecdotal reports from patients, were positive. The full-face approach with BoNTA, used alone or in combination with other aesthetic procedures, is likely to become increasingly common in the coming years.



FIGURE 6 Full-face treatment with onabotulinumtoxinA. Static and dynamic photographs of a 44-y-old man before treatment (A–E) and at 3 wk after treatment (F–J) with a total of 130 units of onabotulinumtoxinA in the glabella, crow's feet, forehead, and platysma. The impact on the platysma (pretreatment, image D; post-treatment, image I) is particularly notable. The result is a younger looking patient, but with preservation of natural expressiveness

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CONFLICT OF INTEREST

The authors declare no potential conflicts of interest with respect to the research, authorship, and publication of this article.

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